

Section Id :	64065338428
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	45
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065381130
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 1 Question Id : 640653566871 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531894488. ✔ YES

6406531894489. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065381131
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 2 Question Id : 640653566872 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

In which of the following scenarios a facility layout problem is relevant?

Options :

6406531894490. ✔ A new facility is added

6406531894491. ✔ A new product is introduced

6406531894492. ✔ A new service is introduced

6406531894493. ✔ Shift in technology

Question Number : 3 Question Id : 640653566873 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Select the disadvantage(s) of a product layout from the following .

Options :

6406531894494. ✔ Lack of flexibility

6406531894495. ✔ High risk, if a machine breaks down then entire line stops

6406531894496. ✖ High work-in-process

6406531894497. ✖ Need specialized supervision

Question Number : 4 Question Id : 640653566875 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

What are the inputs for an order-up-to-level calculation?

Options :

6406531894502. ✔ Sales forecast

6406531894503. ✔ Lead time

6406531894504. ✔ On-hand inventory

6406531894505. ✔ Safety stock

Question Number : 5 Question Id : 640653566878 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

What is Digitalization? Choose all that are applicable

Options :

6406531894515. ✖ Pure Analog to Digital Conversion

6406531894516. ✔ Use of digital technologies to change the processes

6406531894517. ✔ Process of moving towards a digital business

6406531894518.

✖ Mere transformation – encoded in digital format

Question Number : 6 Question Id : 640653566880 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true? Choose all that are applicable

Options :

6406531894523. ✔ Euclidean distance measures distance as the straight line between 2 points

6406531894524. ✔ Metropolitan distance measures distance with respect to the road availability

6406531894525. ✖ None of these

Question Number : 7 Question Id : 640653566882 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?

Options :

6406531894529. ✖ Subtour elimination

6406531894530. ✔ Any capacity restrictions

6406531894531. ✔ A specific starting point

6406531894532. ✖ Distance traveled restriction

Sub-Section Number : 3

Sub-Section Id : 64065381132

Question Shuffling Allowed : Yes

Is Section Default? :

null

Question Number : 8 Question Id : 640653566874 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What brought the first industrial revolution?

Options :

6406531894498. ✔ Mechanization

6406531894499. ✖ Electric Power

6406531894500. ✖ Computer

6406531894501. ✖ Robotics

Question Number : 9 Question Id : 640653566876 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is not a part of the 7 principles of IoT:

Options :

6406531894506. ✖ Big Analog Data

6406531894507. ✔ Quality Control

6406531894508. ✖ Constant connection

6406531894509. ✖ Real time

6406531894510. ✖ Spectrum of insights

Question Number : 10 Question Id : 640653566881 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is the decision variable X_{ij} denotes in a Traveling Salesman Problem (TSP)

Options :

6406531894526. ✖ If the sales visits node-j from node-i (binary)

6406531894527. ✔ If the sales visits node-j immediately after visiting node-i (binary)

6406531894528. ✖ The quantity of an item transferred between node-i and node-j (positive,integer)

Sub-Section Number :	4
Sub-Section Id :	64065381133
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 11 Question Id : 640653566877 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is Digitization ?

Options :

6406531894511. ✔ Conversion of physical 2D drawings to CAD drawings

6406531894512. ✖ Creating PLC logic in a microprocessor-based system to process data and identify insights

6406531894513. ✖ Providing real-time maintenance schedule based on real-time machine performance

6406531894514.

✖ None of these

Sub-Section Number : 5
Sub-Section Id : 64065381134
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 12 Question Id : 640653566879 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

At EOQ, the order cost is

Options :

6406531894519. ✖ Minimum

6406531894520. ✖ Lower than the carrying cost

6406531894521. ✔ Equal to the carrying cost

6406531894522. ✖ Higher than the carrying cost

Question Number : 13 Question Id : 640653566893 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets on the neighbouring shelves. He picked a few milk-powder packets. However, these were not part of his list.

He was searching for Good-day biscuits. He could identify only one packet on the shelf. However,

there were a lot of Sun-feast biscuits lying on the next shelf. So, he picked ten Sun-feast packets. Now map the scenarios with the items below:

Product	Scenario
P1) Tea : Coffee	S1) Complementary
P2) Tea: milk-powder packet	S2) Substitution
P3) Good-day biscuits: Sun-feast biscuits	S3) Inventory dependant demand
	S4) Impulse buying

Options :

6406531894548. ✖ P1-S1, P2-S3, P3-S4

6406531894549. ✔ P1-S2, P2-S1, P3-S3

6406531894550. ✖ P1-S2, P2-S1, P3-S4

Sub-Section Number :6

Sub-Section Id :64065381135

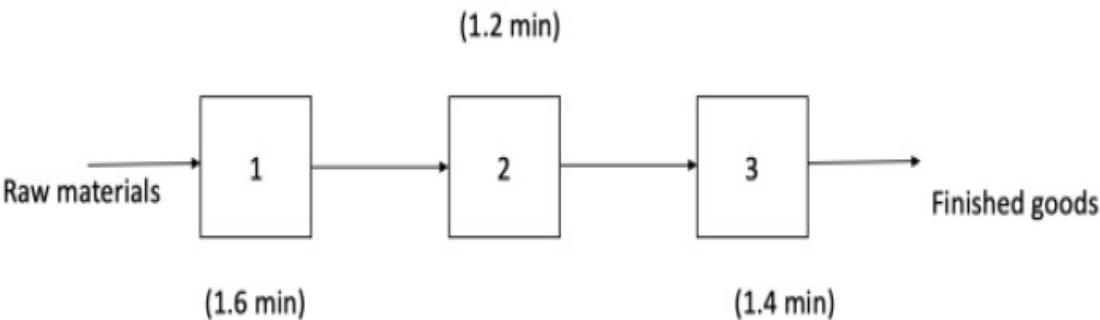
Question Shuffling Allowed :Yes

Is Section Default? :null

Question Number : 14 Question Id : 640653566883 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 3

Question Label : Short Answer Question

A factory has three work centers as shown in the picture below. The time required at each centers are shown in the bracket. Calculate the efficiency of the system (in %).



Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

87 to 88

Sub-Section Number :	7
Sub-Section Id :	64065381136
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 15 Question Id : 640653566884 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Match the following columns by selecting the correct name from Column B for the labels in column A which are depicted in the **‘Figure: EOQ model’** below:

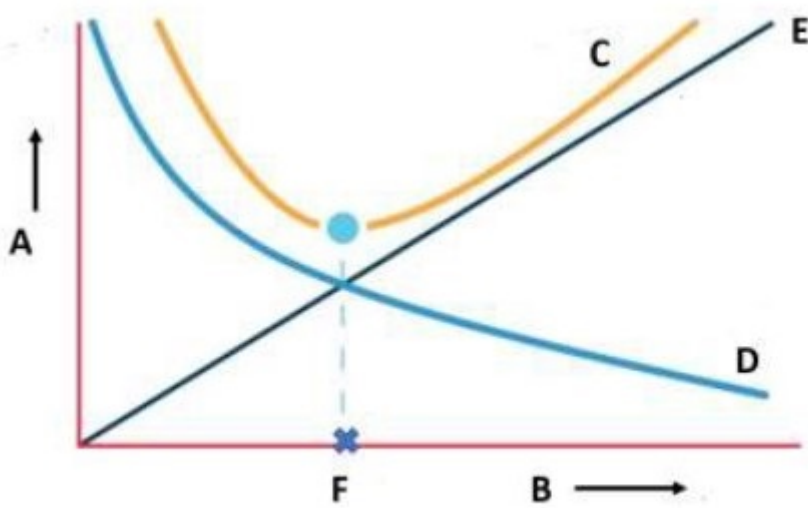


Figure: EOQ model

Column A	Column B
A	Carrying cost
B	Order cost
C	Cost
D	EOQ(Q^*)
E	Quantity
F	Total cost

Options :

6406531894534. ✖ A:Cost; B:Quantity; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q^*)

6406531894535. ✔ A:Cost; B:Quantity; C:Total cost; D:Order cost; E:Carrying cost; F:EOQ(Q^*)

6406531894536. ✖ A:Quantity; B:Cost; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q^*)

6406531894537. ✖ A:Quantity; B:Cost; C:Total Cost; D:Order cost; E:Carrying cost; F:EOQ(Q^*)

6406531894538. ✖ A:Cost; B:EOQ(Q^*); C:Total Cost; D:Order cost; E:Carrying cost; F:Quantity

6406531894539. ✖ A:Total cost; B:EOQ(Q^*); C:Cost; D:Order cost; E:Carrying cost; F:Quantity

6406531894540. ✖ A:Total cost; B:Quantity; C:Cost; D:Carrying cost; E:Order cost; F: EOQ(Q^*)

Sub-Section Number :

8

Sub-Section Id :

64065381137

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 16 Question Id : 640653566885 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Calculate the **Euclidean distance** between two machines located at the coordinates A(5,10) and B (2,6) in a shop floor

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Question Number : 17 Question Id : 640653566886 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Calculate the **Metropolitan** distance between two machines located at the coordinates A(5,10) and B (2,6) in a shop floor

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

6.8 to 7.2

Sub-Section Number :

9

Sub-Section Id :	64065381138
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653566887 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (18 to 21)

Question Label : Comprehension

Ashoke Equipment company estimates its ordering cost at Rs 400 and carrying cost to be Rs. 2/year/unit. The estimated annual requirement is 10,000 units at a unit price of Rs. 10. Calculate the given subquestions:

Sub questions

Question Number : 18 Question Id : 640653566888 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the economic order quantity (EOQ)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 19 Question Id : 640653566889 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the number of orders per year.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 20 Question Id : 640653566890 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Calculate the annual inventory carrying cost

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 21 Question Id : 640653566891 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

While rechecked, Ashoke Equipment company finds that the order cost should have been 20% more than the currently estimated value. What should be the percentage increase in the EOQ in this case? Enter your answer without the '%' symbol.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.4 to 9.6

Sub-Section Number : 10

Sub-Section Id : 64065381139

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 22 Question Id : 640653566892 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Currently, a company is using a layout which consists of 9 workstations. 200 sandwiches are produced in a day on this layout. A typical day consists of a single shift which runs for 10 hours. The sandwich-making process consists of 9 activities (the information is provided in the table below).

Activity ID	Activity Description	Activity Time (in seconds)	Preceding Activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Cutting	20	C
F	Butter applying	5	E
G	Vegetable filling	5	F, E
H	Sandwich baking	15	G
I	Sandwich packing	5	H

What is the current efficiency of the line where each activity is performed on a separate workstation

(Enter only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

11.10 to 11.12

Sub-Section Number : 11

Sub-Section Id : 64065381140

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653566894 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (23 to 24)

Question Label : Comprehension

Suppose a company wants to expand its logistics network and locate a new warehouse within a network of three existing facilities. The coordinates of the existing warehouses are: A(200,50), B(300,100), C(100,150) and the respective outbound goods volumes are 2500 units, 1300 units, 5000 units. Determine the location of the new warehouse (round off the answer to the nearest integer) and answer the given subquestions.

Sub questions

Question Number : 23 Question Id : 640653566895 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Short Answer Question

x-cordinate of the new warehouse: